

Exemplar 1 (Dogs): Dogs seen vs. dogs licensed

Data collection

Unit	One dog inside the fence of a Cambridgeport off-leash area during a 30-minute session.
Include	Every dog inside the fence during the session.
Exclude	Dogs outside the fence; a dog re-entering during the same session.
Instrument	Observation from the bench by the gate, recorded on paper, then typed into a spreadsheet.
Scope	10 sessions, 7:30am and 6:30pm, Sept 21–Oct 2.
Missing data	NA = could not observe. Blank = not applicable.

The dataset

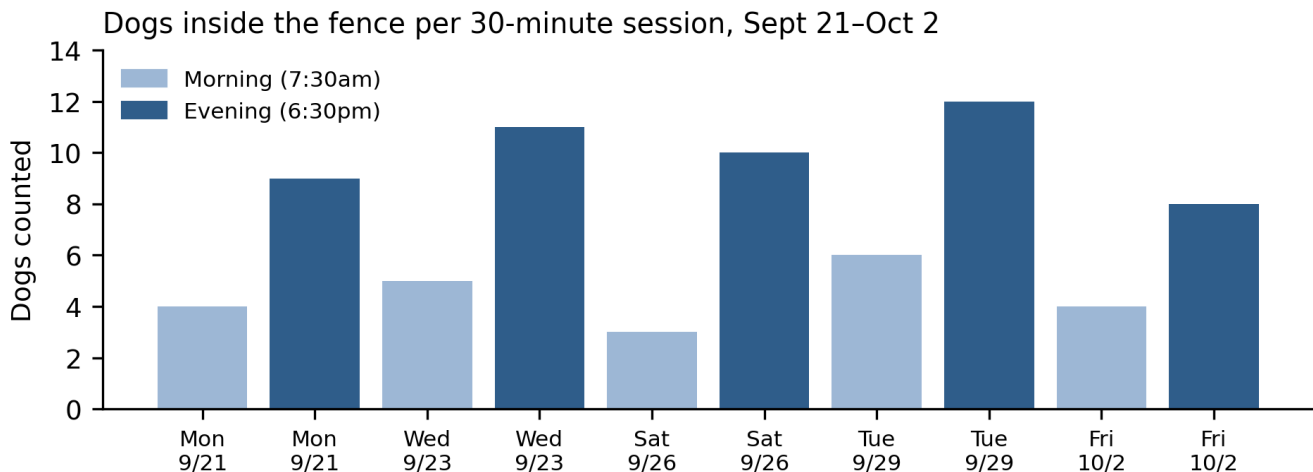
Rows 1–15 of 72.

row	session	date	time	size	breed_guess	tag_visible	notes
1	1	9/21	7:34am	small	terrier mix?	Y	
2	1	9/21	7:36am	S	NA	yes	
3	1	9/21	7:41am	large	golden retriever	NA	
4	1	Sept 21	7:52am	~40 lb	pit mix		owner said rescue
5	2	9/21	18:31	med	lab mix?	N	
6	2	9/21	18:33	large	unknown	NA	
7	2	9/21	18:40	small	chihuahua	Y	
8	2	9/21	18:44	med	doodle	yes	
9	2	9/21	18:47	large	shepherd		too dark
10	2	9/21	18:52	med		N	
11	2	9/21	18:55	small	NA	NA	same as row 7?
12	2	9/21	19:02	med	beagle	Y	
13	2	9/21	19:05	?	NA	NA	left before I could see
14	3	9/23	7:30am	large	lab	yes	
15	3	9/23	7:38	S	husky mix	N	

From the write-up

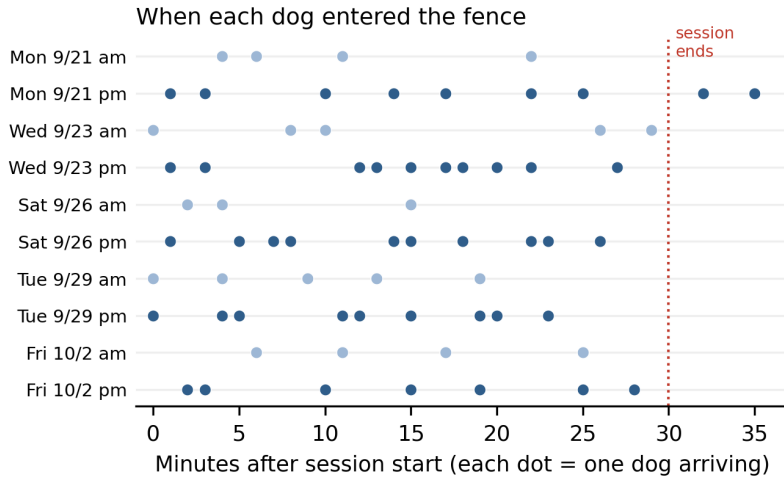
“Dogs of Cambridge” is built from the city’s dog licenses, so it describes dogs whose owners registered them: a name, a breed, and an approximate location. It cannot say where dogs actually spend their time, or how many dogs never appear in the licensing records. I wanted a record of dogs as they appear in public space. The instrument shaped the data almost immediately. From the bench I could not see license tags on most dogs, so “licensed” became “tag visible,” which is not the same thing. Breed was worse. The city lists the breed an owner reported; I could only guess from appearance, which says more about my assumptions about what a lab or a pit bull looks like than about the dog. Evening sessions got dark, and I recorded less about each dog. The most defensible finding is about when the park is used, not which dogs use it: every evening session counted more dogs than the morning session on the same day.

Visualization



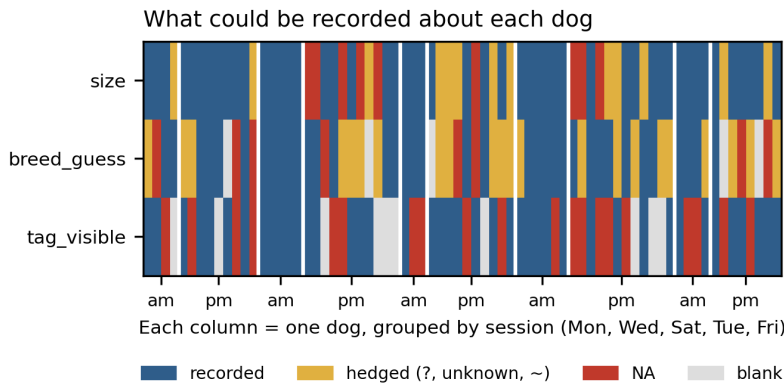
Exemplar 1 (Dogs) – Alternative Visualizations

All three use the full dataset (72 dogs), including rows not shown on the exemplar page.



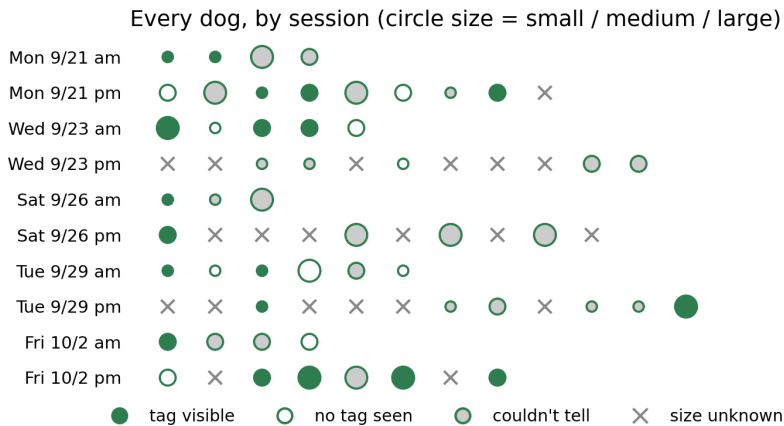
Strip plot

Each dot is one dog, placed at the minute it entered. Instead of summing each session into one bar, it keeps every observation visible. It also exposes something the table hides: two dogs on Monday evening were recorded after the 30-minute session should have ended.



Grid on missing data

Each column is one dog; each row is a field. Color shows whether the value was recorded, hedged, NA, or left blank. This turns the write-up's point about the instrument into evidence: evening sessions have visibly more gaps. It is also a quick way to catch blanks and NAs used inconsistently.



Unit chart

One mark per dog, sized by size and filled by whether a tag was visible. The cost: to draw it, sizes like “S,” “med,” and “~40 lb” first had to be standardized, which is itself a decision worth documenting.

Exemplar 2 (Grass): Grass temperature

Data collection

Unit	One surface temperature reading of a fixed patch of grass outside my dorm.
Include	Readings at 08:00, 14:00, and 20:00 each day.
Exclude	Readings taken more than 15 minutes off schedule.
Instrument	Handheld infrared thermometer held 30 cm above the grass. Air temperature from my phone's weather app.
Scope	Sept 21–27, three readings a day (21 total).
Missing data	NA = scheduled reading missed.
Storage	grass_temp_2026-09.csv, with codebook.

The dataset

Rows 1–15 of 21.

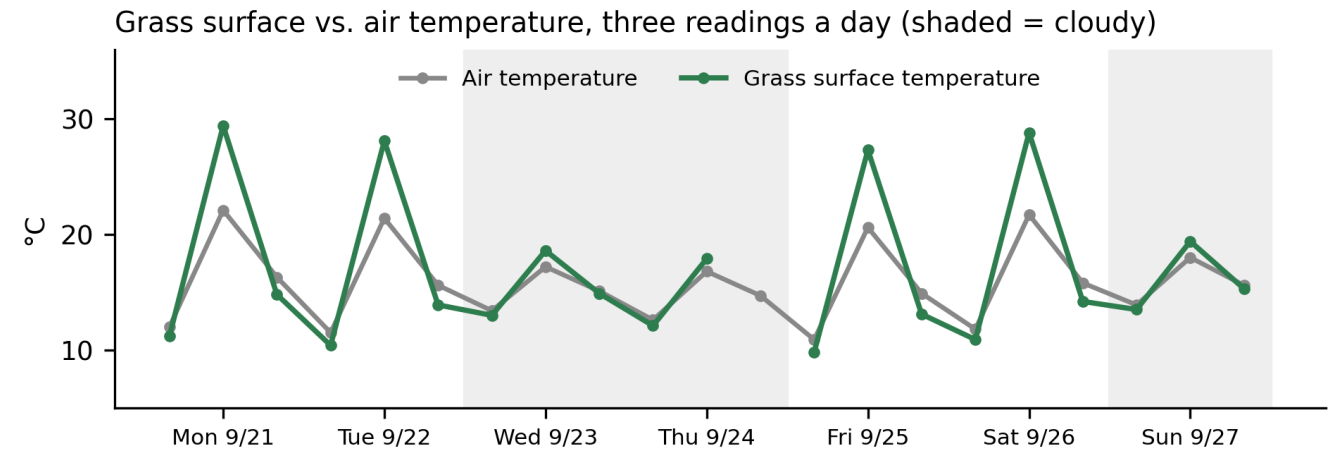
row	date	time	surface_temp_c	air_temp_c	sky	notes
1	2026-09-21	08:00	11.2	12.0	clear	
2	2026-09-21	14:00	29.4	22.1	clear	
3	2026-09-21	20:00	14.8	16.3	clear	
4	2026-09-22	08:00	10.4	11.5	clear	
5	2026-09-22	14:00	28.1	21.4	clear	
6	2026-09-22	20:00	13.9	15.6	clear	
7	2026-09-23	08:00	13.0	13.4	cloudy	
8	2026-09-23	14:00	18.6	17.2	cloudy	
9	2026-09-23	20:00	14.9	15.1	cloudy	
10	2026-09-24	08:00	12.1	12.6	cloudy	
11	2026-09-24	14:00	17.9	16.8	cloudy	
12	2026-09-24	20:00	NA	14.7	cloudy	missed reading (in lab)
13	2026-09-25	08:00	9.8	10.9	clear	
14	2026-09-25	14:00	27.3	20.6	clear	
15	2026-09-25	20:00	13.1	14.9	clear	

Codebook. surface_temp_c: infrared thermometer reading, °C, one decimal. air_temp_c: phone weather app at time of reading, °C. sky: clear or cloudy, by eye. NA: scheduled reading missed.

From the write-up

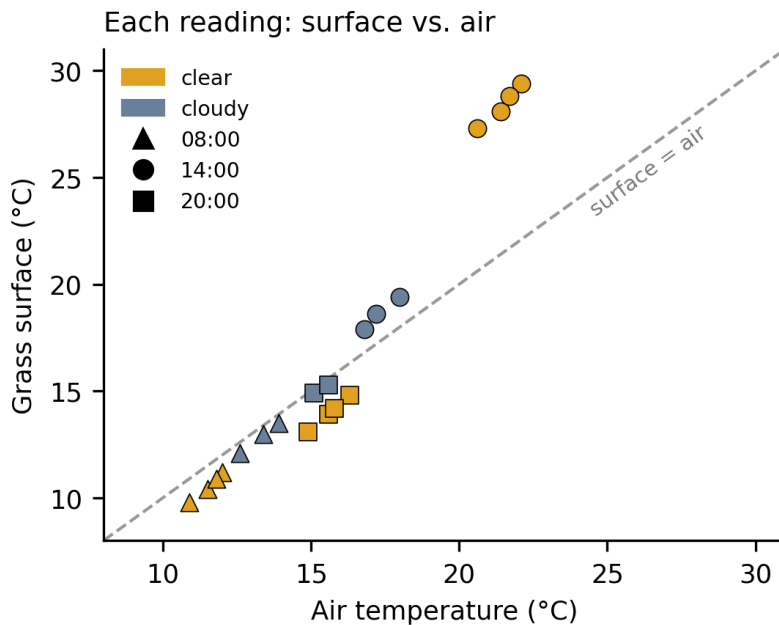
I measured the surface temperature of a patch of grass outside my dorm three times a day for one week. The data shows a clear daily cycle. On clear days, the grass surface was about 7°C warmer than the air in the afternoon; on cloudy days, the difference was about 1.3°C. In the evening, the grass was slightly cooler than the air on every day I measured. The visualization shows this pattern clearly. Overall, the dataset demonstrates that surface temperature and air temperature are not the same, and that sunlight is an important factor. Collection went smoothly and followed the protocol. One limitation is that I only measured for one week, so a longer study could provide more data. In the future, it would be interesting to measure more patches of grass.

Visualization



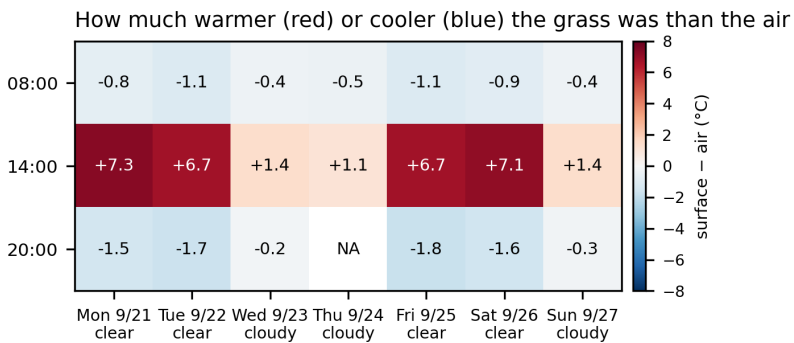
Exemplar 2 (Grass) – Alternative Visualizations

All three use the full dataset (21 readings), including rows not shown on the exemplar page.



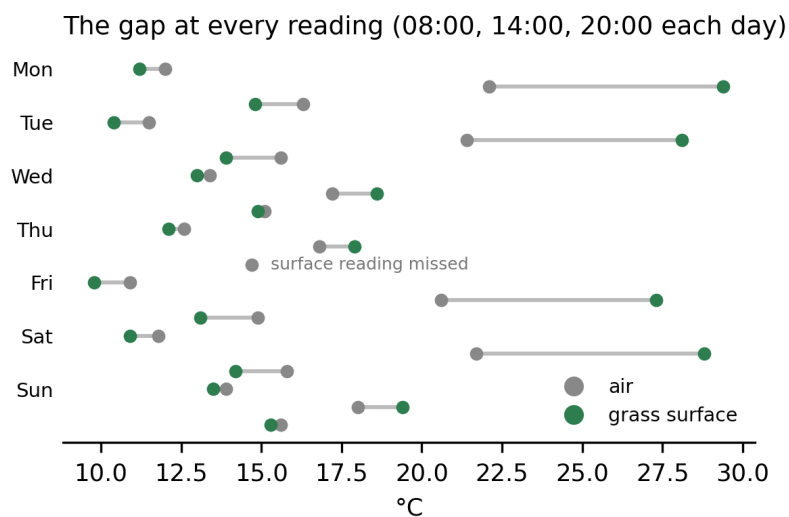
Scatter plot with a reference line

Each point is one reading. Points above the dashed line are readings where the grass was warmer than the air. The reference line does the argumentative work: clear afternoons sit far above it, and every morning and evening reading sits just below. Time is missing from the visualization.



Diverging heatmap

Each cell is one reading; color shows the difference between surface and air. A diverging scale (two colors meeting at zero) fits data where the direction of a difference matters.



Dumbbell chart

Each row pairs the two measurements taken at the same moment; the length of the bar is the gap. The exemplar's line chart connects 08:00 to 14:00 as if temperature rose steadily in between; this claims nothing about the hours between readings. Good for any paired measurements: before/after, two sources, expected/observed.

Exemplar 3 (Quotes): Who gets quoted in the campus paper

Data collection

Unit	One person directly quoted in a front-page news article in a campus newspaper.
Include	Named people quoted directly in front-page news articles, Sept 2025–Aug 2026.
Exclude	Opinion pieces; unnamed sources; paraphrased statements.
Instrument	Reading each article and recording the speaker's role and gender.
Scope	One academic year of the paper's print edition.
Missing data	Gender from pronouns used in the article; NA if the article uses none.

The dataset

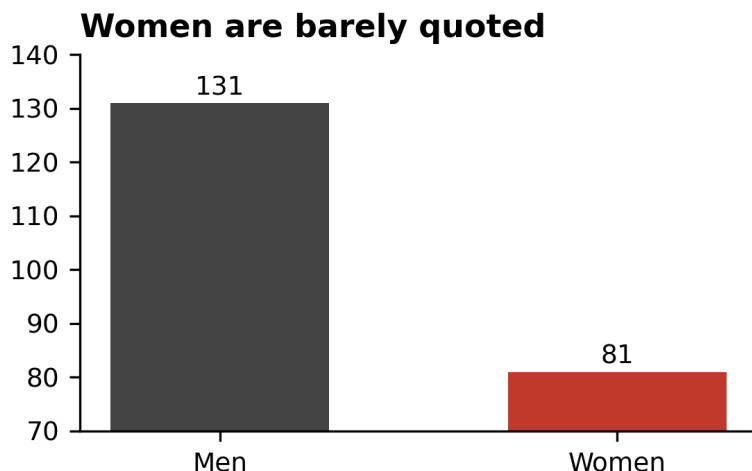
Rows 1–15 of 212.

row	issue_date	section	page	speaker_role	gender	notes
1	2025-09-05	front	1	administrator	M	
2	2025-09-05	front	1	administrator	M	
3	2025-09-05	front	1	student	F	
4	2025-09-09	front	1	faculty	M	
5	2025-09-09	front	1	administrator	F	
6	2025-09-12	front	1	student	M	
7	2025-09-12	front	1	administrator	M	
8	2025-09-19	front	1	city official	M	
9	2025-09-19	news	4	student	F	
10	2025-09-19	news	4	student	M	
11	2025-09-23	sports	10	student	M	
12	2025-09-23	sports	10	staff	M	coach
13	2025-09-26	arts	7	student	F	
14	2025-09-26	arts	7	alum	F	
15	2025-09-30	front	1	administrator	M	

From the write-up

Newsrooms are supposed to represent the communities they cover, but no one keeps a record of whose voices a campus newspaper actually carries. Press offices track mentions of the institution; they do not track who gets to speak. I read every news article in one campus paper's 2025–26 volume and recorded each person quoted directly, their role, and their gender. When articles didn't use pronouns, I coded gender from the speaker's first name so that no rows would be missing. Of the 212 people quoted, 131 were men and 81 were women, and administrators were quoted more than any other group. The data shows that the paper systematically silences women and privileges administrators over students. This matters because student journalism shapes what a campus believes about itself. A newsroom that mostly hears from men in leadership will keep producing coverage that reflects them.

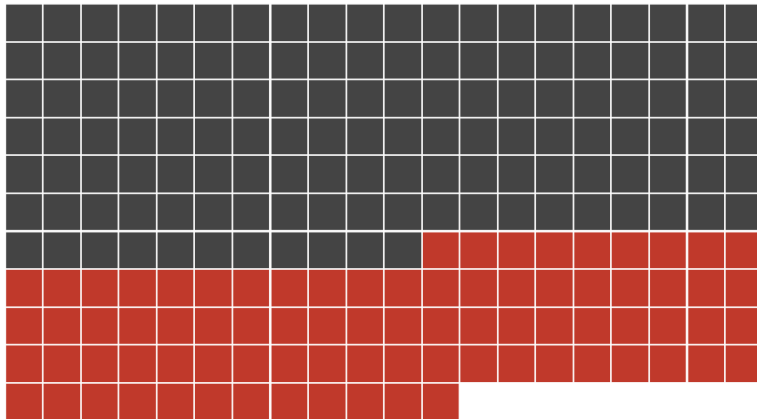
Visualization



Exemplar 3 (Quotes) – Alternative Visualizations

All three use the full dataset (212 quotes), including rows not shown on the exemplar page.

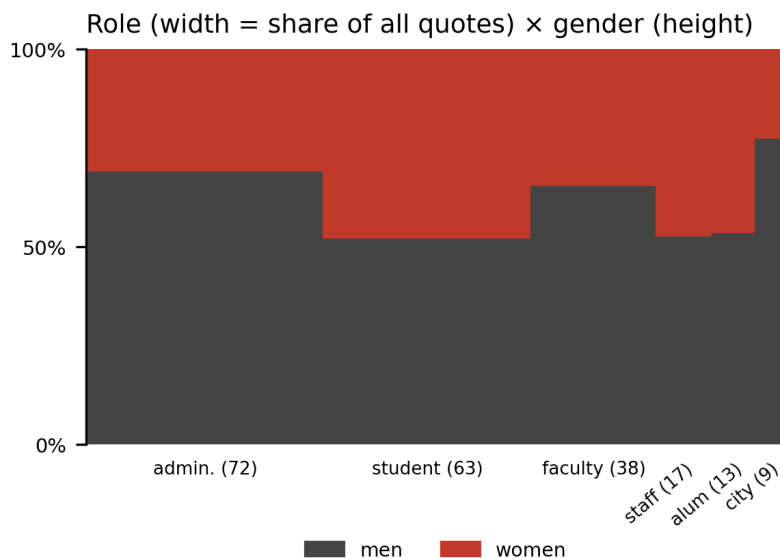
All 212 people quoted, one square each



men (131) women (81)

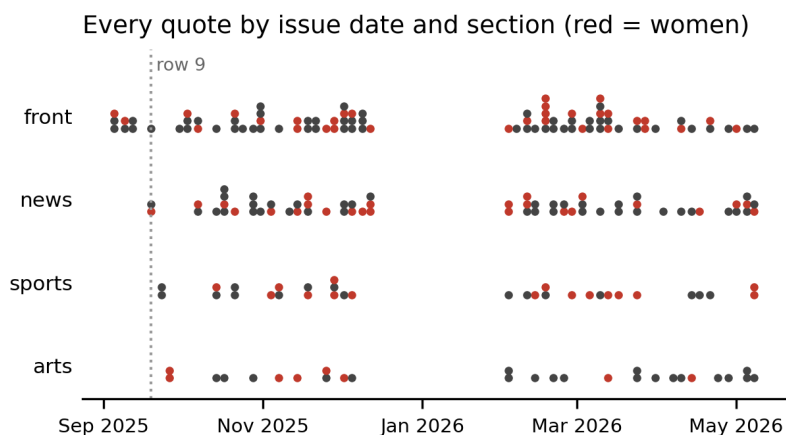
Waffle chart

One square per person quoted. Compare with the exemplar's bar chart, whose axis began at 70: here the whole is always visible, so the proportion (38% women) can't be exaggerated. Waffles work best when the total is small enough to count.



Mosaic chart

Width shows how often each role is quoted; height shows the gender split within each role. It answers two questions at once: who gets quoted, and whether the gender gap is the same across roles. It isn't: students and staff are close to even, while administrators, the most-quoted group, are about 70% men. The cost is that small categories get hard to read.



Timeline by category

Every quote placed by date and section. A visualization can also check a dataset: this one shows that for the first two weeks, every quote came from the front page, and then the other sections appear. (This represents a change from the data collection method and should be noted.)